Kris Ko

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Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science in Computer Engineering, Honours — *Sep. 2025 – May 2030*

Technical Skills

Languages: C, C++, Python, ARM Assembly, Verilog
Embedded: Bare-metal drivers, DMA, interrupts, RTOS scheduling and synchronization, FreeRTOS, CMSIS-DSP
Controls: PID control, Kalman filtering, sensor fusion
Microcontrollers: ARM Cortex-M (STM32F4/L5/H7), ESP32
Interfaces: I2C, SPI, CAN, UART, GPIO, PWM, ADC, Ethernet, UDP
Tools: Git, GDB, JTAG/SWD, STM32CubeIDE, CMake, Make, GoogleTest, logic analyzer, oscilloscope

Experience

Embedded Flight Software Engineering Co-op — May 2026 – Aug. 2026
Waterloo Aerial Robotics Group — *Waterloo, ON*

- Developed quadrotor support for ZeroPilot, a **FreeRTOS-based C++ flight controller on STM32H7**, integrating firmware, build tooling, CI, and AirSim software-in-the-loop tests to validate control logic before flight.
- Implemented STABILIZE mode **cascaded PID attitude control** with outer attitude and inner rate loops, motor mixing with desaturation that preserves attitude authority at high throttle, and a **DMA-driven DShot ESC driver**; flight tested and tuned across 10+ flights.
- Built a timestamped **dual ICM42688 IMU driver** over a shared SPI bus using a **ping-pong DMA state machine** that merges both FIFO streams into one ordered timeline; root-caused a delta-time bug feeding the Mahony filter that was causing significant attitude lag.
- Ported and optimized an **AHRS EKF** to CMSIS-DSP, cutting execution from 20ms to 340µs to meet its 1 kHz budget, verified with a custom RTOS profiler; added an FFT-based **harmonic notch filter** that tracks the dominant motor vibration frequency and suppresses its harmonics in the gyro signal.
- Eliminated control task stalls from blocking FatFS SD-card writes by moving logging behind a **message queue**, where producer threads push to the queue and a dedicated **low-priority preemptible thread** absorbs the blocking writes.
- Designed a **4-state altitude Kalman filter** (accelerometer propagation, barometer/GPS correction) with innovation gating and covariance-inflation recovery to restore correction after sustained measurement rejection, feeding a **three-loop cascaded altitude controller** (position/velocity/acceleration at 25/100/200 Hz) with LiDAR terrain following; validated in SITL.
- Wrote DMA-based drivers: a UART u-blox GPS driver (UBX configuration with NMEA fallback, full u-blox M-series support) and I2C drivers for ICP20100 barometer, INA228 power module, and TF02-Pro LiDAR rangefinder.

Software Lead — Sep. 2022 – May 2025
MATE ROV Competition — *Halifax, NS*

- **Led 10+ students** developing software for an underwater ROV; built an Ethernet UDP thruster control system.

Projects

KoRTOS | *C, ARM Assembly, Cortex-M4* — Dec. 2025 – Present

- Built a **preemptive, priority-based RTOS kernel** from scratch on STM32F446RE with round-robin among equal-priority tasks on a 1 ms SysTick; architected with a **swappable port layer**, isolating all ARM assembly and register access, so a new architecture requires no changes to the scheduling core.
- Implemented **counting semaphores, mutexes with chained priority inheritance, and ISR-safe message queues**, each with its own wait list, selectable FIFO/priority unblock policies, and timeouts; measured worst case priority inversion blocking at **20.3ms without inheritance vs 257µs with** for a 20 ms interfering task, flat across 1–20 ms workloads with inheritance.
- Designed **deferred context switching via PendSV** with split MSP/PSP stacks and **naked assembly functions** for manual register and stack management, preserving EXC_RETURN across switches.
- Wrote **register-level GPIO/SPI/I2C/USART/CAN HAL** (polling + interrupt modes; configurable CAN bit timing and hardware filtering) and integrated CAN with the kernel via **ISR-to-queue RX handoff** and mutex-protected TX mailboxes.
- Authored the linker script (VMA/LMA separation), startup code with full vector table and .data/.bss init, and a Makefile producing separately linkable kernel/HAL static libraries; verified the kernel and HAL with 22 sample apps.